

ACCESS CONTROL LIST

An **Access Control List (ACL)** is a set of rules that is used to filter network traffic and control access to and from a network. ACLs are used on network devices such as routers, switches, and firewalls to permit or deny traffic based on various criteria, such as IP addresses, protocols, port numbers, and more.

There are two types of ACL:

1. Standard ACL (**Range:** 1–99 and 1300–1999)
2. Extended ACL (**Range:** 100–199, 2000–2699)
3. Named ACLs: No numeric range (alphanumeric names)

Purpose of ACLs:

ACLs are mainly used for the following purposes:

Traffic Filtering: ACLs filter incoming and outgoing traffic on a network based on predefined rules, helping to control which types of traffic are allowed or denied.

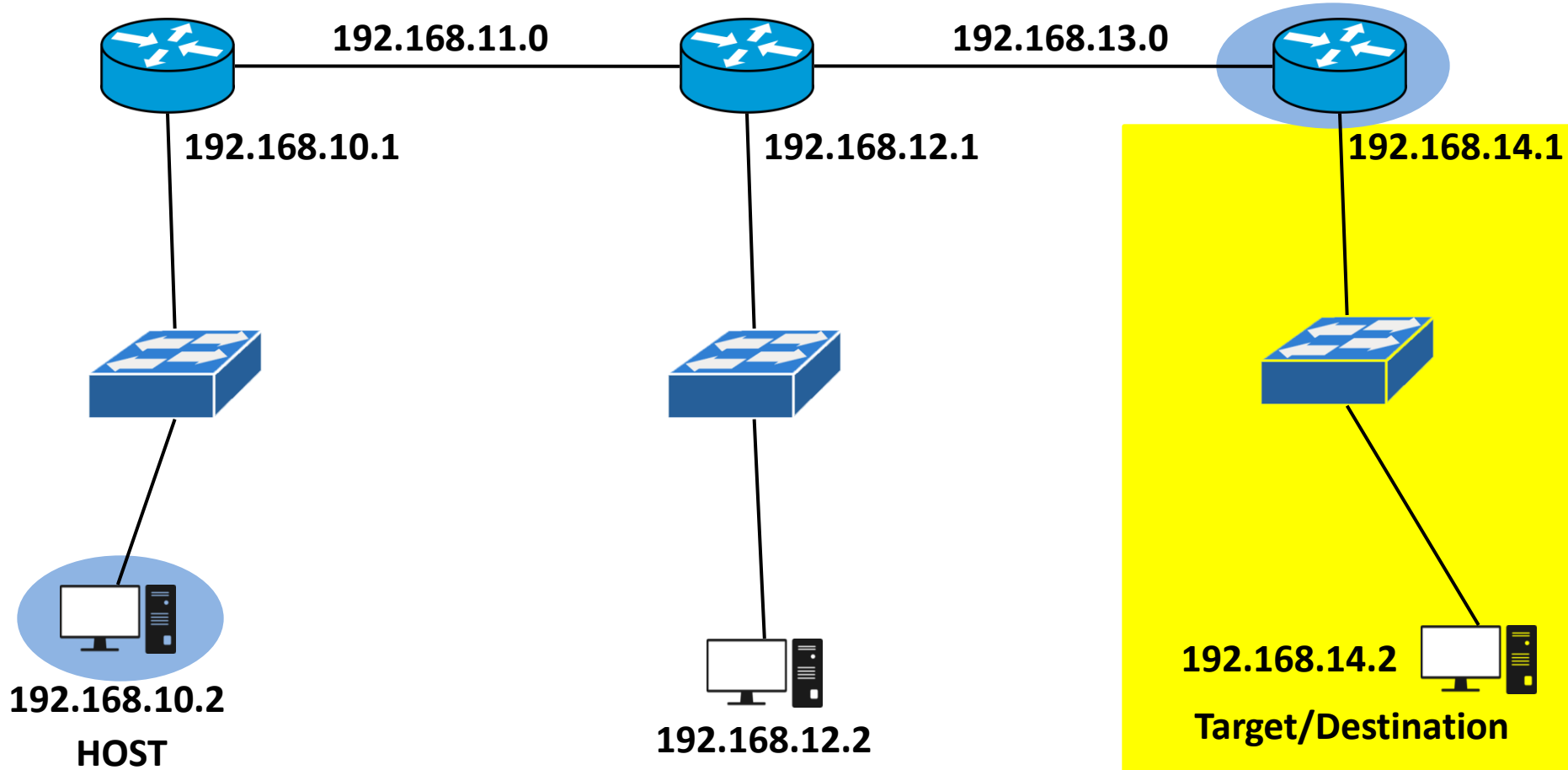
Network Security: ACLs provide a security mechanism to restrict access to specific network resources by permitting or denying traffic from certain IP addresses or networks.

Access Control: ACLs are used to enforce security policies by allowing only authorized users or devices to access specific parts of a network.

Quality of Service (QoS): ACLs can be used to prioritize traffic for better performance and to control bandwidth usage.

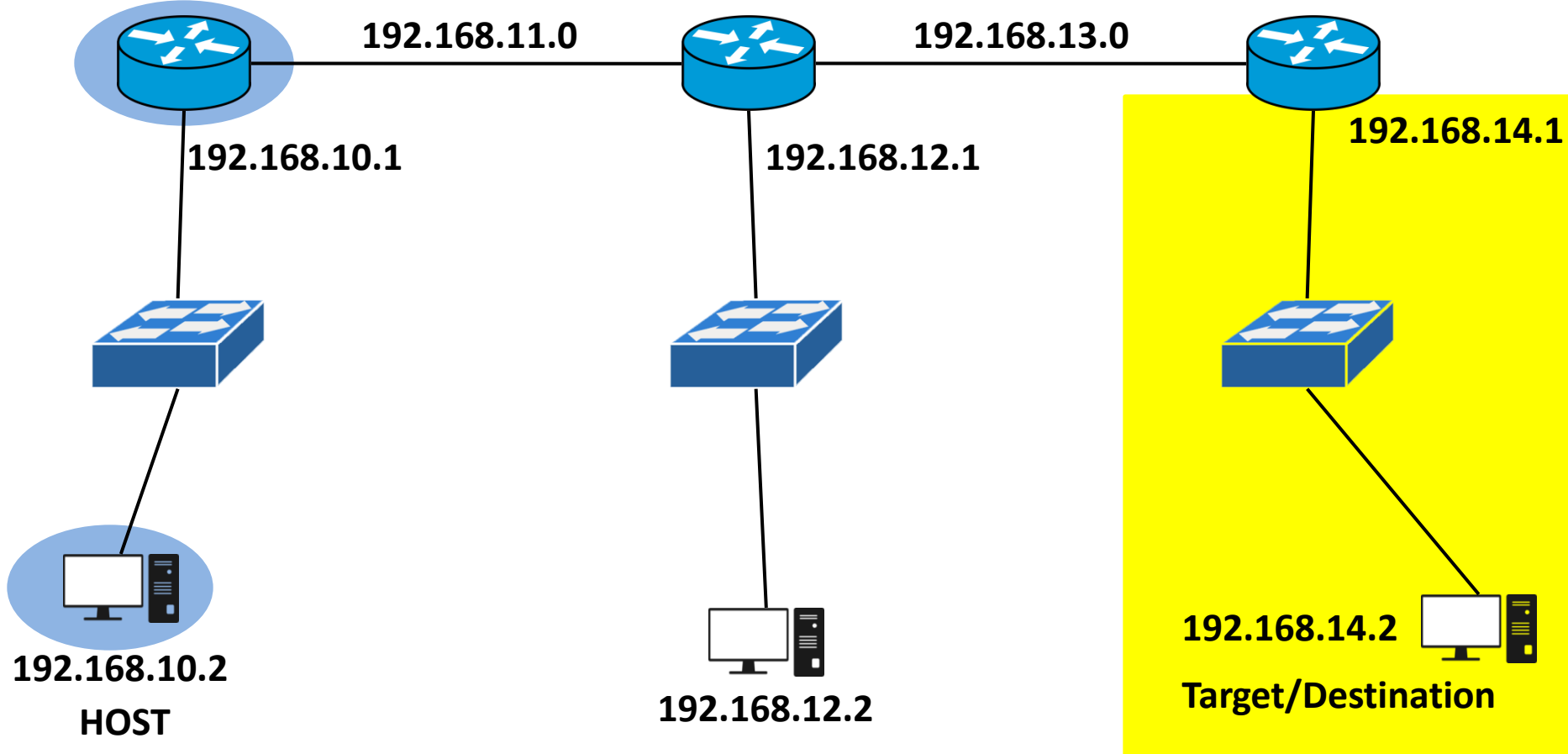
Standard ACL:

Configured on node Closest to the Target/Destination



Extended ACL:

Closest to the HOST/Source



Extended ACLs:

- Extended ACLs provide more granular control by filtering traffic based on multiple criteria, such as source and destination IP addresses, protocols (TCP, UDP, ICMP), and port numbers.
- They allow for more specific traffic filtering and are commonly used in more complex network environments.
- Configured Closest to the HOST/Source.
- Extended ACL: (**Range:** 100–199, 2000–2699)
- Named ACLs No numeric range (alphanumeric names)
- Source and Destination IP/TCP/UDP must be defined separately in Extended.

Configuration Example of Standard ACL:

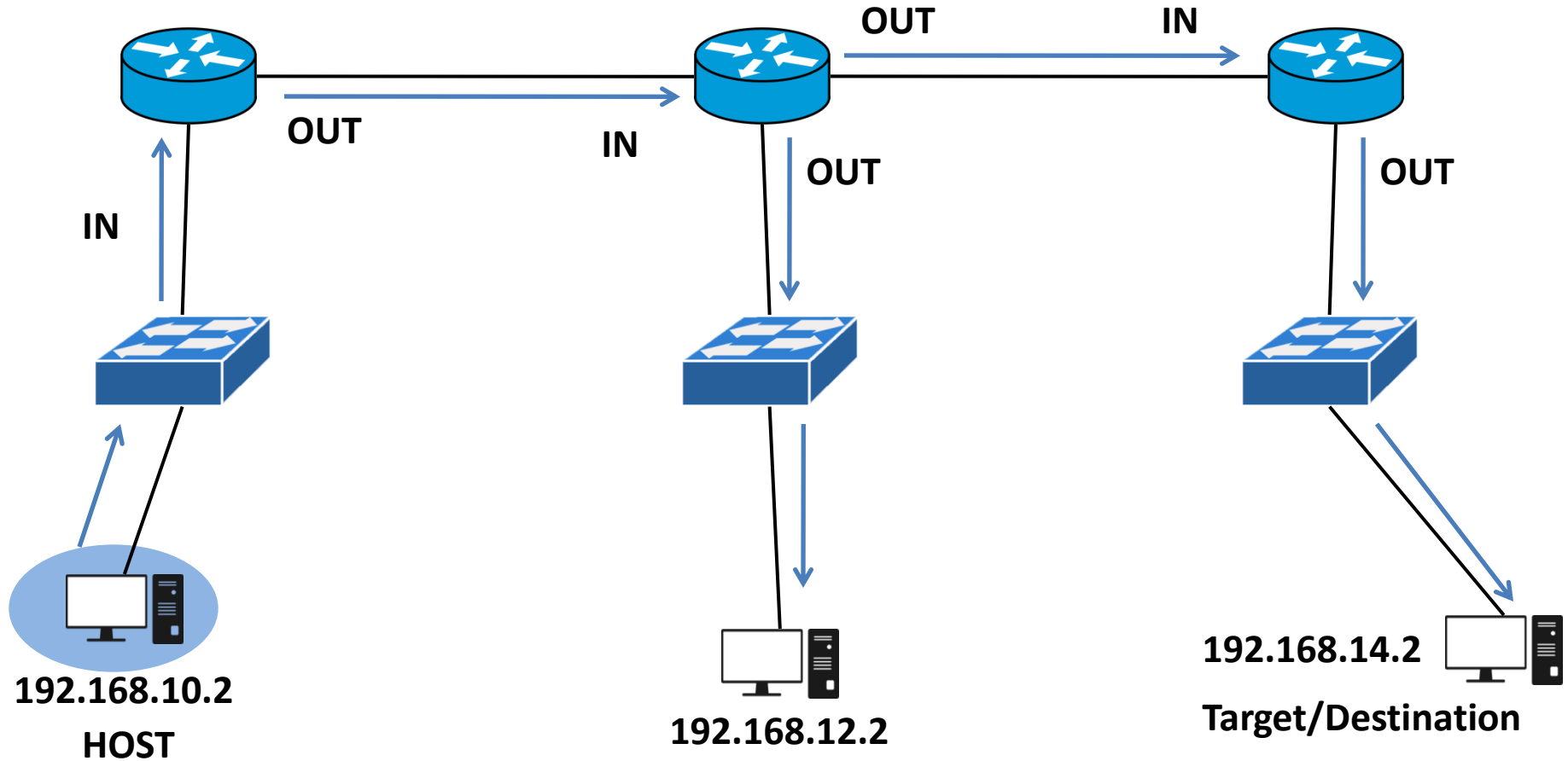
Block All and Permit Single user/network:

```
Router(config)#access-list 100 permit tcp  
192.168.1.0 0.0.0.255 172.16.0.0 0.0.255.255 eq 80  
Router(config) access-list 100 deny ip any any
```

Allow All and Block Single user/network:

```
Router(config)#access-list 100 deny ip 192.168.1.0  
0.0.0.255 172.16.0.0 0.0.255.255 eq 80  
Router(config) access-list 100 permit ip any any
```

How to decide inbound or Outbound:



Applying ACLs:

Inbound ACLs: Applied to incoming traffic on an interface

Router(config)#interface GigabitEthernet0/0

Router(config)#ip access-group 100 in

Outbound ACLs: Applied to outgoing traffic on an interface.

Router(config)# interface GigabitEthernet0/1

Router(config)# ip access-group 100 out